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1. Project Title

We are going to develop and design industry level **AI Resume Screening System using NLP, Machine Learning, and Computer Vision (OCR technique)**.

2. Introduction

Artificial Intelligence (AI) has significantly improved human–computer interaction, with applications such as chatbots and automated decision-making systems widely used in domains like healthcare, education, and recruitment. These systems enhance efficiency by automating tasks and enabling faster decisions, but traditional approaches often lack deep contextual understanding.

In recruitment, manual resume screening is time-consuming and inconsistent. To address this, advanced technologies like Natural Language Processing (NLP), Machine Learning (ML), and Computer Vision (CV) are used to process structured and unstructured data more effectively. This project focuses on developing a Resume Screening System that automates resume evaluation by extracting and analyzing content using NLP, and matching it with job descriptions using ML algorithms.

Additionally, OCR and Computer Vision are used to process scanned resumes, ensuring comprehensive data extraction. The system improves recruitment efficiency by providing accurate candidate ranking, reducing manual effort, and ensuring consistency, demonstrating the practical application of AI in real-world scenarios.

2.1 Objectives

The main objectives of these projects are..

1. To develop an intelligent Resume Screening System using AI techniques.
2. To automate the extraction of useful information from resumes using NLP.
3. To compare resumes with job descriptions using Machine Learning algorithms.
4. To generate a matching score and rank candidates accordingly.
5. To process both text-based and scanned resumes using OCR and Computer Vision.
6. To reduce manual effort and improve efficiency in the recruitment process.
7. To ensure accurate and consistent evaluation of candidate profiles.
8. To store and manage data efficiently using a relational database (SQLite).

3. Project Category

This project falls under the category of **Artificial Intelligence (AI)** and **Machine Learning (ML)**. It also involves concepts of **Natural Language Processing (NLP)** and **Computer Vision (CV)**. The project focuses on developing an intelligent resume screening system that can analyze resumes, extract relevant information, and match them with job descriptions. It is a software-based application with intelligent features aimed at improving the recruitment process.

4. Tools / Platform

The development of the Resume Screening System requires various software tools and platforms. These tools are used for resume processing, NLP analysis, machine learning, and system implementation.

4.1 Software Platform

- **Operating System:** Windows / Linux
- **Programming Language:** Python
- **Frontend Technology:** Streamlit
- **Database:** SQLite
- **Libraries:** Pandas, NumPy, NLTK / spaCy
- **Machine Learning Library:** Scikit-learn
- **Computer Vision & OCR:** OpenCV, Tesseract OCR

4.2 Hardware Platform

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Hard Disk:** Minimum 100 GB

4.3 Tools

- **IDE:** VS Code / Jupyter Notebook
- **Web Browser:** Google Chrome / Mozilla Firefox
- **Dataset:** Sample resume dataset / job description dataset

5 . Problem Statement

the recruitment process in many organizations involves manually screening a large number of resumes, which is time-consuming, inefficient, and prone to human bias. Traditional methods often fail to accurately match candidate profiles with job requirements due to the lack of intelligent analysis of unstructured resume data.

Additionally, many systems are unable to process scanned or image-based resumes effectively, leading to loss of important candidate information.

There is a need for an automated and intelligent system that can efficiently extract, analyze, and evaluate resume data using advanced technologies. The system should be capable of understanding textual content through Natural Language Processing (NLP), performing accurate matching using Machine Learning (ML) algorithms, and handling diverse resume formats using Computer Vision (CV) and Optical Character Recognition (OCR).

Therefore, this project aims to develop a Resume Screening System that improves the efficiency, accuracy, and consistency of the recruitment process by automating resume analysis and candidate-job matching

5.1 Project Description

The Resume Screening System is an intelligent application designed to automate and improve the evaluation of candidate resumes using Artificial Intelligence. It integrates Natural Language Processing (NLP), Machine Learning (ML), and Computer Vision (CV) to analyze both structured and unstructured data efficiently.

The system allows recruiters to upload resumes and job descriptions in formats like PDF, DOCX, or scanned images. Text-based resumes are processed directly, while image-based resumes are converted into text using OCR techniques.

After extraction, NLP methods such as tokenization, stopword removal, and lemmatization are applied. The data is then transformed into numerical form using techniques like TF-IDF, and Machine Learning algorithms are used to calculate a matching score between resumes and job descriptions.

Candidates are ranked based on these scores, and the system provides options to shortlist candidates and generate reports. All data is stored in a SQLite database for efficient management. The system is implemented in Python using tools like Pandas, Scikit-learn, and Streamlit, ensuring scalability and easy future enhancements.

Overall, the system reduces manual effort, improves accuracy, and offers an efficient approach to resume screening using AI technologies.

6. Scope of the Work

The scope of this project focuses on the design and development of an intelligent Resume Screening System that automates the analysis and evaluation of candidate resumes using Artificial Intelligence techniques. The system handles key functionalities such as resume processing, candidate-job matching, and result generation within a local environment. Users can upload resumes and job

descriptions in formats like PDF, DOCX, and scanned images. Text is extracted using NLP techniques, while OCR and Computer Vision are used for image-based resumes. The extracted data is processed to identify key features such as skills, experience, and relevant keywords. Machine Learning algorithms are used to compare resumes with job descriptions and generate matching scores. Based on these scores, candidates are ranked, and the system allows users to view results, shortlist candidates, and generate reports. All data is stored in a SQLite database for efficient management. The system is developed using Python and libraries like Pandas and Scikit-learn, with a Streamlit-based interface. The system is modular, allowing future enhancements such as integration with job portals or cloud deployment. However, the current scope is limited to a prototype and does not include real-time deployment, large-scale databases, or advanced deep learning models.

6.1 Advantage of this Project

1. The system automates resume screening, reducing the need for manual effort by recruiters.
2. It saves time by quickly processing and evaluating a large number of resumes.
3. It improves accuracy by using NLP and Machine Learning to match candidate profiles with job requirements.
4. It effectively handles unstructured data and extracts important information such as skills and experience.
5. It supports multiple resume formats including PDF, DOCX, and scanned images using OCR and Computer Vision.
6. It ranks candidates based on matching scores, helping recruiters identify suitable candidates easily.
7. It reduces human bias by relying on automated and objective evaluation methods.
8. It provides efficient data management through structured storage and easy retrieval using SQLite.
9. The system has a scalable and modular design, allowing future enhancements like cloud integration and advanced models.

7. Analysis -1. DFD 2. E-R Diagram 3. Use Case Diagram

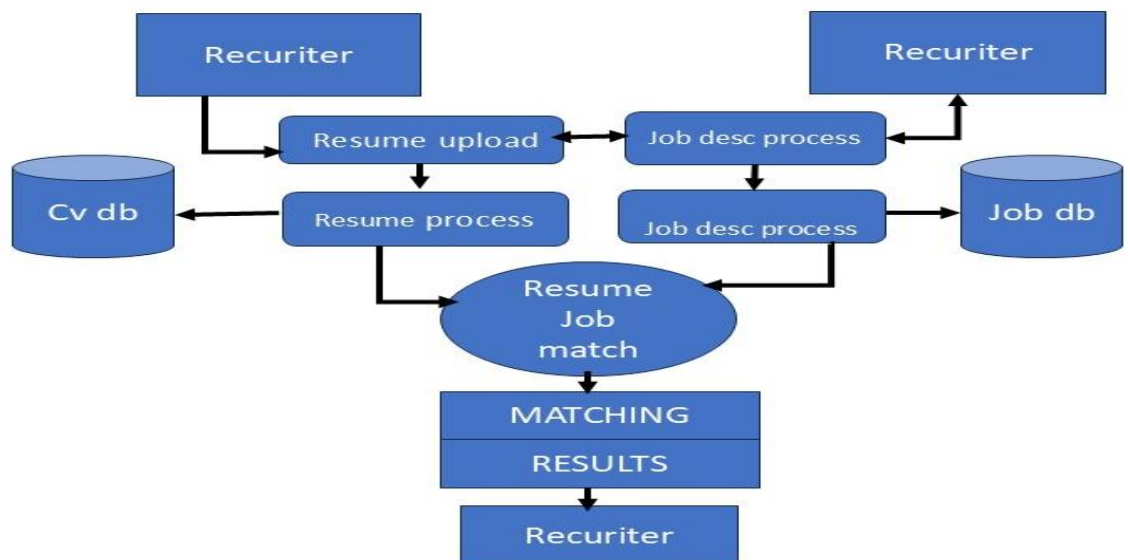
The analysis phase is an important part of the project, as it helps in understanding the system requirements, functionality, and overall workflow of the Resume Screening System. Existing recruitment systems are studied to identify their limitations, particularly the time-consuming manual screening process and lack of intelligent analysis of resumes. Based on the requirements, the proposed system is defined. The system analysis includes identifying functional and non-functional requirements. Functional requirements involve uploading resumes and job descriptions, extracting text from different formats, processing data using Natural Language Processing (NLP), performing matching using Machine Learning (ML), and generating ranked results. Non-functional requirements include system performance, accuracy, security, scalability, and usability. During analysis, different system components such as the user interface, resume parsing module, NLP processing module, machine learning matching module, computer vision and OCR module, and database are identified. The interaction and data flow between these components are also analyzed to ensure smooth and efficient system operation. Various diagrams such as Data Flow Diagram (DFD), Entity Relationship (ER) Diagram, and Use Case Diagram are used to represent the system structure and workflow clearly. This analysis helps in designing an efficient system architecture and serves as a foundation for successful implementation of the Resume Screening System.

7.1 Data Flow Diagram (DFD)

The Data Flow Diagram represents the flow of data within the Resume Screening System. It shows how data moves from the user to the system and how it is processed.

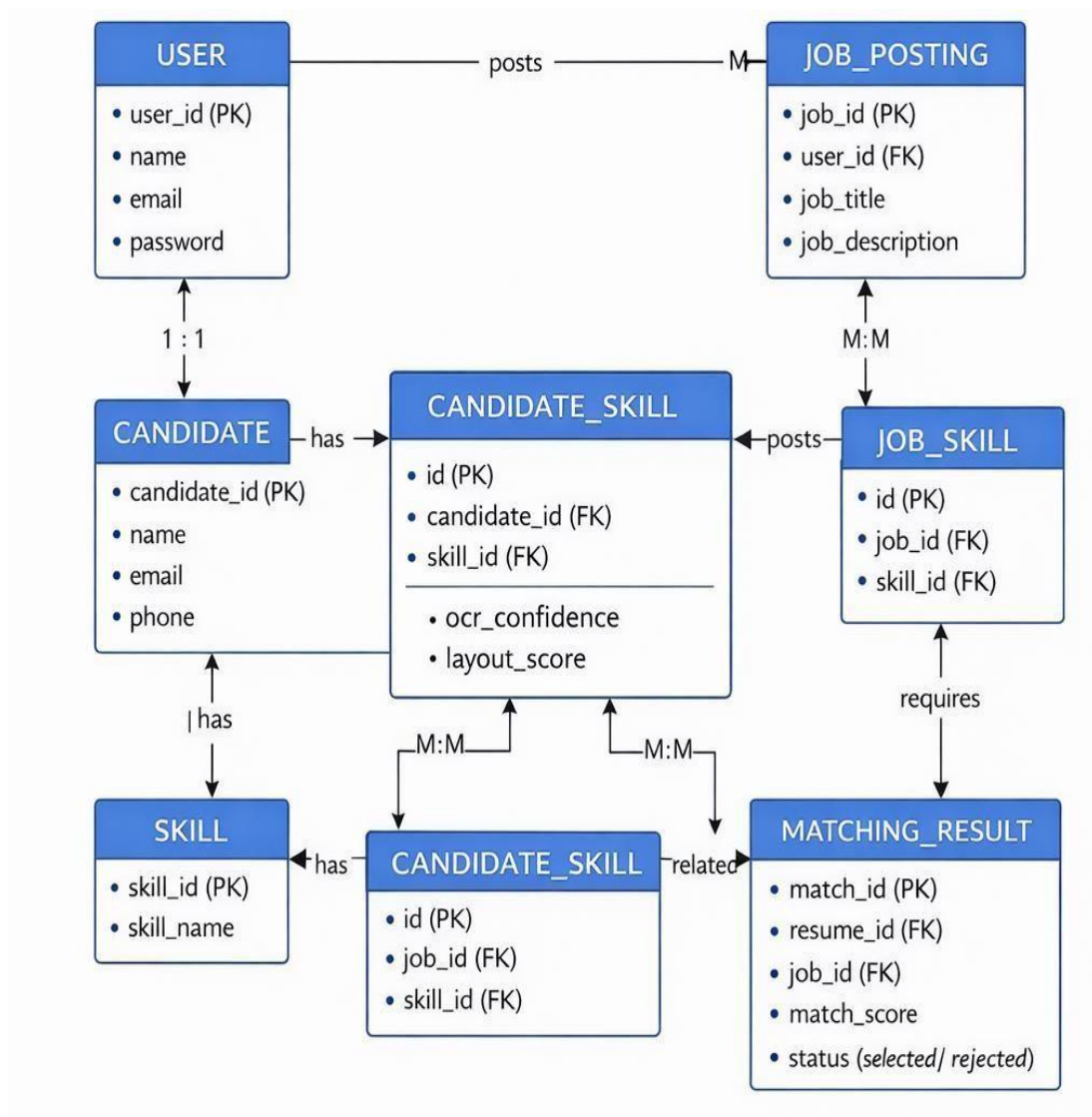
- In **Level 0 (Context Diagram)**, the user uploads resumes and job descriptions, and the system returns matching results.
- In **Level 1**, the system is divided into processes such as resume upload, text extraction, NLP processing, and resume-job matching.
- In **Level 2**, detailed processes like OCR for scanned resumes, feature extraction, and similarity calculation are shown.

The DFD helps in understanding how data is processed step-by-step inside the system.

LEVEL 0 DFD**LEVEL 1 DFD****LEVEL 2 (NLP+OCR+ML)**

7.2 E -R Diagram

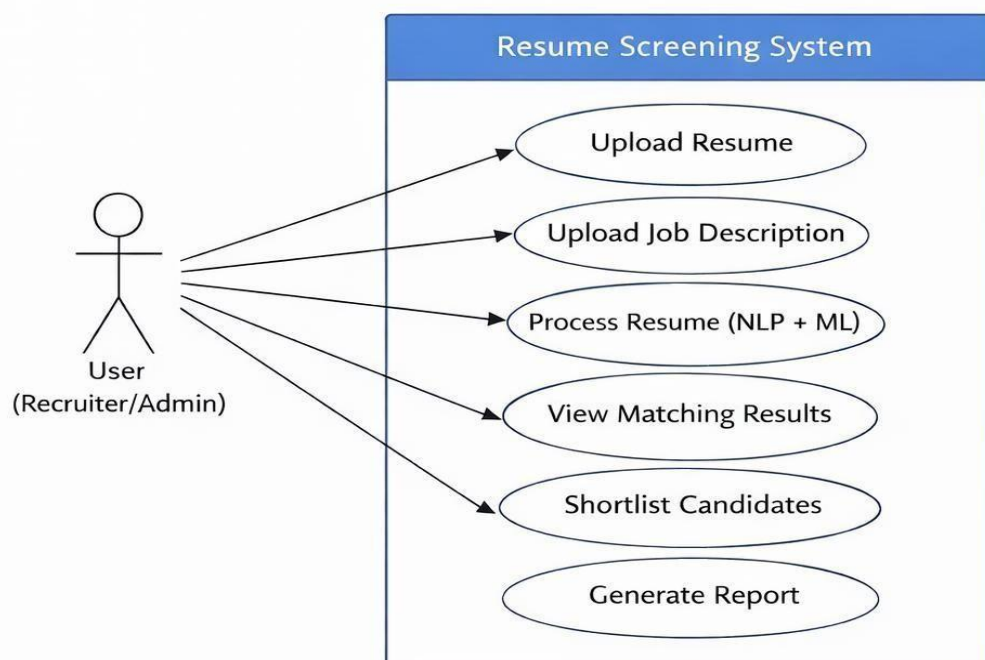
The ER diagram represents the logical structure of the database used in the Resume Screening System. It defines entities such as User, Candidate, Resume, Job Posting, Skill, and Matching Result along with their relationships. The design ensures proper data storage, retrieval, and management using a relational database (SQLite)



The E-R Diagram represents logical structure of used in **AI RESUME SCREENING SYSTEM**

7.3 Use case Diagram

The Use Case Diagram illustrates the interaction between the user (recruiter/admin) and the Resume Screening System. It highlights the primary functionalities of the system, including uploading resumes, uploading job descriptions, processing resumes, viewing matching results, shortlisting candidates, and generating reports. The diagram helps in understanding the functional behavior and scope of the system from the user's perspective.



The Use Case diagram visually represents the interactions between the user and the Resume Screening System.

8 . COMPLETE DATABASE AND TABEL

The Resume Screening System uses SQLite as the database management system. SQLite is a lightweight, file-based relational database that is easy to manage and suitable for small to medium-scale applications. The database stores structured information such as user details, candidate data, resumes, job postings, skills, and matching results in organized tables. Python libraries such as SQLite3 are used to interact with the database, while Pandas is used for data processing and analysis.

8.1 USER Table

Field Name	Data Type	Description
user_id	INTEGER (PK)	Unique ID for user
name	TEXT	Name of recruiter/admin
email	TEXT	User email
password	TEXT	User password

8.2 CANDIDATE Table

Field Name	Data Type	Description
candidate_id	INTEGER (PK)	Unique ID for candidate
name	TEXT	Candidate name
email	TEXT	Candidate email
phone	TEXT	Contact number

8.3 SKILL Table

Field Name	Data Type	Description
skill_id	INTEGER (PK)	Unique skill ID
skill_name	TEXT	Skill name

8.4 Candidate_skill Table

Field Name	Data Type	Description
id	INTEGER (PK)	Unique ID
candidate_id	INTERGER(FK)	Links for candidate
skill_id	INTEGER(Fk0	Links to skill

8.5 RESUME Tabel

Field Name	Data Type	Description
resume_id	INTEGER (PK)	Unique resume ID
Candidate_id	INTEGER(FK)	Links to candidate
resume_text	TEXT	Extracted resume content
resume_type	TEXT	Text/scanned
upload_date	DATA	Resume upload data
ocr_confidence	REAL	OCR accuracy score
layout_score	REAL	Resume format score

8.6 JOB_SKILL Table

Field Name	Data Type	Description
job_id	INTEGER (PK)	Unique ID
Job_skill	INTEGER(FK)	Links to job
Skill_id	INTEGER(FK)	Links to skill

8.8 MATCHING_RESULT Table

Field Name	Data Type	Description
match_id	INTEGER (PK)	Unique match ID
resume_id	INTEGER(FK)	Links to resume
job_id	INTEGER(FK)	Links to job
match_score	REAL	Similarity score
status	TEXT	Selected/Rejected

9. COMPLETE STRUCTURE

The Resume Screening System is designed using a modular structure to ensure proper organization and efficient functioning. The system integrates Natural Language Processing (NLP), Machine Learning (ML), and Computer Vision (CV) techniques. It consists of frontend, backend, database, and resume processing components working together. The frontend provides a user interface where the user (recruiter/admin) can upload resumes and job descriptions and view the results. The backend handles the core processing logic, including resume parsing, NLP-based text preprocessing, and machine learning-based matching.

It also manages communication between different modules of the system. The **database (SQLite)** stores structured information such as user details, candidate information, resumes, job postings, and matching results. This ensures efficient data storage, retrieval, and management. The resume processing module includes text extraction from documents and **OCR-based** processing for scanned resumes using Computer Vision techniques. This module ensures that both text-based and image-based resumes are handled effectively. This modular and structured approach makes the system easy to maintain, scalable, and adaptable for future enhancements such as cloud deployment and integration with real-time recruitment platforms.

9.1 NUMBER OF MODULES

1. User Management Module

Handles user login and authentication for recruiters/admins. It ensures secure access to the system.

2. Resume Upload & Parsing Module

Allows users to upload resumes in PDF, DOCX, or image formats. It extracts text from resumes using parsing techniques.

3. NLP Processing Module

Processes the extracted text using Natural Language Processing techniques such as tokenization, stopword removal, and lemmatization.

4. Machine Learning Matching Module

Compares resume content with job descriptions using ML algorithms (e.g., TF-IDF and similarity measures) and generates a matching score.

5. Computer Vision & OCR Module

Handles scanned or image-based resumes using OpenCV and OCR to extract text and ensure compatibility with different formats.

6.Database Module

Stores user details, candidate data, resumes, job postings, and matching results using SQLite.

7.Result & Reporting Module

Displays ranked candidates, allows shortlisting, and generates reports for decision-making.

9.2 Data Structure

The system uses a relational database structure implemented in SQLite. Data is organized into multiple tables with defined relationships to ensure efficient storage and retrieval.

Main data structures include:

- **User Table** – Stores recruiter/admin details such as name, email, and password.
- **Candidate Table** – Stores candidate information such as name, email, and contact details.
- **Resume Table** – Stores resume content, type (text/scanned), and extracted information.
- **Job Posting Table** – Stores job titles and job descriptions.
- **Skill Table** – Stores different skills required for jobs and possessed by candidates.
- **Matching Result Table** – Stores similarity scores and selection status.

Primary keys and foreign keys are used to maintain relationships between tables. SQLite is used for data storage, while Python libraries such as Pandas are used for data processing and manipulation.

9.3 Implementation methodology

The system is developed using a structured approach to ensure proper integration of all components.

Step 1: Requirement Analysis

Project requirements were analyzed to identify key functionalities such as resume upload, text extraction, **NLP processing**, job matching, and data storage.

Step 2: System Design

Diagrams like DFD, ER Diagram, and Use Case Diagram were created to define system structure and workflow.

Step 3: Database Design

SQLite database tables were designed to store candidate details, resumes, job descriptions, and matching results.

Step 4: Resume Processing & NLP

Text extraction techniques were implemented, followed by **NLP methods** such as tokenization, stopwords removal, and lemmatization.

Step 5: Machine Learning Development

TF-IDF and similarity measures were used to compare resumes with job descriptions and generate matching scores.

Step 6: Computer Vision & OCR

OCR and image processing techniques were applied to convert scanned resumes into readable text.

Step 7: Backend Development

Python was used to implement core logic including **NLP processing**, **ML matching**, and database operations.

Step 8: Frontend Development

A user interface was developed using Streamlit for uploading resumes, entering job descriptions, and viewing results.

Step 9: Testing and Validation

The system was tested for accuracy, functionality, **OCR performance**, and data handling to ensure reliability.

9.4 List of Reports

the system generates and maintains the following reports:

1. User Activity Report

Contains details of users such as login information and activity history.

2. Candidate Report

Stores candidate information including name, contact details, and uploaded resumes.

3. Resume Analysis Report

Provides extracted resume data, resume type (text or scanned), and NLPprocessed information.

4. Matching Result Report

Showing matching scores between resumes and job descriptions along with candidate ranking and selection status.

5. Job Posting Report

Maintains records of job titles, descriptions, and requirements.

6. System Performance Report

Tracks processing time, matching accuracy, and overall system efficiency.

10 Overall Network Architecture

The Resume Screening System follows a modular client-server-like architecture implemented in a local environment. The system is designed to ensure smooth interaction between the user interface, backend processing modules, machine learning components, and the database. It enables efficient handling of resume data, processing, and result generation.

10.1 Architecture Overview

1. Client Side (Frontend Layer)

The client side provides a user interface developed using Streamlit.

- Users (recruiters/admins) interact with the system through a web browser.
- The interface allows users to upload resumes and job descriptions.
- It displays matching results, rankings, and reports.
- User inputs are sent to the backend for processing.

2.Application layer

The backend is implemented using Python scripts.

- Handles core system logic and workflow.
- Manages resume parsing, NLP processing, and ML-based matching.
- Coordinates communication between different modules.
- Interacts with the database for storing and retrieving data.

3. NLP, Machine Learning & Computer Vision Layer

- NLP techniques are used for text preprocessing (tokenization, stopword removal, lemmatization).
- Machine Learning models (TF-IDF and similarity measures) are used to compare resumes with job descriptions and generate matching scores.

- Computer Vision and OCR are used to process scanned or image-based resumes and convert them into machine-readable text

4. Database Layer (SQLite)

- Stores user data, candidate details, resumes, job postings, and matching results.
- Maintains relational structure using primary and foreign keys.
- Ensures efficient data storage, retrieval, and management.

11. Implementation of Security Mechanisms at Various Levels

Security is an important aspect of the Resume Screening System. Since the system handles sensitive information such as user details, candidate resumes, and job-related data, proper security mechanisms are implemented at different levels to ensure data protection and safe system operation.

The security mechanisms are implemented at the following levels:

User Authentication Level

- Basic authentication is implemented for user access (recruiter/admin).
- Users must log in before accessing system functionalities.
- Passwords are stored in hashed format rather than plain text.
- Session handling ensures that only authorized users can access system data.

This prevents unauthorized access to the system.

Application-Level Security

- Input validation is applied to resume uploads and job descriptions.
- File type validation ensures only valid formats (PDF, DOCX, images) are accepted.
- Python-based processing reduces risks such as SQL injection.
- Error handling mechanisms are implemented to prevent system crashes and data leaks.

These measures ensure secure application-level operations.

Database Level Security

- SQLite database is stored locally in a secure environment.
- Access to the database file is restricted to authorized users only.

- Primary and foreign keys are used to maintain data integrity.
- Sensitive information such as user credentials is stored securely using hashing techniques.

This protects stored data from unauthorized access and modification.

Network Level Security

- The system runs on localhost (127.0.0.1) during development.
- External access is restricted to prevent unauthorized connections.
- For future deployment, HTTPS can be used to secure communication.

This ensures safe data transmission over the network.

Machine Learning & NLP Security

- **ML** models and NLP processing scripts are stored securely within the backend.
- Only backend processes can access model files and processing logic.
- No direct user access is provided to internal processing modules.
- Input data is sanitized before being processed by ML/NLP components.

This prevents misuse and ensures reliable model performance.

Computer Vision & OCR Security

- Uploaded image files are validated before processing.
- OCR operations are performed securely within the system environment.
- No external execution of uploaded files is allowed.

This ensures safe handling of scanned or image-based resumes.

12. Future Scope and Enhancement

The Resume Screening System has been successfully developed using NLP, Machine Learning, and Computer Vision techniques for automated resume analysis and candidate-job matching. However, there are several possibilities for future improvements and enhancements.

Future Scope

1. Cloud Deployment

The system can be deployed on cloud platforms such as AWS, Azure, or Google Cloud to make it accessible to organizations globally.

2. Advanced Deep Learning Models

Advanced models such as BERT or Transformer-based architectures can be used to improve semantic understanding and matching accuracy.

3. Integration with Job Portals

The system can be integrated with platforms like LinkedIn or job portals to automatically fetch resumes and job postings.

4. Resume Recommendation System

The system can suggest suitable job roles to candidates based on their skills and experience.

5. Multilingual Resume Processing

Support for multiple languages can be added to handle resumes from different regions.

6. Real-Time Recruitment Analytics

Advanced dashboards and analytics can be integrated to provide insights into hiring trends and candidate performance.

Further Enhancement

- Integration of advanced skill extraction using Named Entity Recognition (NER).
- Implementation of explainable AI to justify matching decisions.
- Performance optimization for handling large datasets.
- Role-based access control for better system management.

12.1 Conclusion

The **Resume Screening System** project demonstrates the integration of **Natural Language Processing (NLP)**, **Machine Learning (ML)**, and **Computer Vision (CV)** into a single intelligent system.

The system successfully: